

RESEARCH ARTICLE



Does collective citizen input impact government service provision? Evidence from SeeClickFix requests

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Abstract

Does collective citizen input impact government priorities and performance in service provision? As cities increasingly offer interactive issue reporting options through online platforms and mobile apps, I investigate whether comments and follows on requests drive faster issue resolution. I theorize that this input signals issue validity, severity, or scrutiny, assisting city administrators in prioritizing and responding to requests. Leveraging a novel dataset of requests from 100 cities, I find that comments and follows double the probability of request closure and that collaborative requests are resolved up to 5 days faster on average than non-collaborative requests. By comparing two cities that use the same platform but that differ in the observability of citizen collaboration, I isolate a distinct and significant influence of collective citizen input on government responsiveness. The findings speak to how technological advances enable information-sharing from citizens that can shape service delivery rules and outcomes.

Evidence for practice

- New information technology, including interactive platforms for submitting service requests, can enable citizen-to-citizen interactions in coproduction.
- These technological advances can also shape service delivery rules and outcomes when collective citizen input on requests, contributed during the request handling process, signals issue validity, severity, or scrutiny to city administrators.
- Collaborative service requests with comments and follows are more likely to be completed and are completed faster than non-collaborative service requests.
- Given these benefits of citizen-to-citizen collaboration in coproduction, attention should be devoted to potential equity gaps in not only service request submission, but also service request collaboration.

INTRODUCTION

Coproduction creates normative value for society, through citizen participation in democratic governance, as well as instrumental value for government, through improvements to information-sharing and efficiency in applications such as public service delivery (Nabatchi et al., 2017). Opportunities for coproduction are shaped by the institutional, informational, and technological environments in which citizens live and governments operate. Technological advances impact coproduction by, for

example, reducing costs that can lower some barriers to citizen participation and by allowing for easier, reciprocal information-sharing between citizens and government. Scholars of coproduction have begun to consider these impacts, focusing on the ways that the Internet, social media, smartphones, and apps structure and enable interactions amongst citizens and between collectives of citizens and government (Chatfield & Reddick, 2018; Clark et al., 2013; Kim et al., 2022; Linders, 2012; Meijer, 2011).

In particular, online platforms and mobile apps for *interactive* service request reporting now enable citizens

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to contribute information to government *during* a service request's lifespan, such as through follows and comments on neighbors' requests, with the potential to influence how requests are addressed and resolved. While prior work has examined citizen satisfaction and feedback after services have been completed or has highlighted the monitoring and accountability features of online platforms (Allen et al., 2020; Kelly & Swindell, 2002; Linders, 2012), this study explores a crowdsourcing function earlier in the service delivery process and examines whether information produced by citizen collaboration impacts how public services are delivered. That is, the paper seeks to answer the following question: Does collective citizen input—specifically, information produced by citizen-to-citizen “collaboration” on service requests—impact government priorities and responsiveness in service provision?

I construct a novel panel of service request data to explore whether collective citizen input matters for how requests are completed, specifically how quickly issues are resolved. I identify “collaborative” service requests as those with comments and follows, creating and signaling a group of concerned citizens. I find that collaboration on requests doubles the probability of request closure and that requests with comments and follows are resolved up to 5 days faster on average than non-collaborative requests. Recognizing that requests with collaborators may be different from those without in ways that affect response times, I employ a difference-in-differences approach to better account for the problem of unobserved heterogeneity. By comparing response times across two cities that use the same request submission platform but that differ in the ability of city administrators to observe comments and follows, I isolate a distinct and significant influence of collaboration on government responses.

Moreover, interviews with city administrators provide evidence that the prioritization of collaborative requests is driven by signals of validity, severity, and scrutiny that are communicated through comments and follows on requests. Though these mechanisms are not separately distinguishable given the research design of the current study, I use request heterogeneity to further explore the importance of severity signals. Overall, the results suggest that citizen-to-citizen interactions in coproduction may lead to larger impacts on government priorities and performance than individual coproduction, at least in the commissioning phase of the service cycle. Given these benefits of citizen-to-citizen collaboration in coproduction, attention should be devoted to potential equity gaps in not only service request submission, but also service request collaboration.

SERVICE PROVISION AND COPRODUCTION IN THE DIGITAL AGE

Service provision is a central role of local government, but time and resource constraints frequently lead cities to

turn to residents for assistance. The logic is that “supplementing—or perhaps supplanting—the labors of paid public officials with the service-directed activities of urban dwellers” can “raise both the quality and the efficiency of municipal services” (Brudney & England, 1983). In this way, residents serve as volunteers or watchdogs to aid the work of public officials. Yet, the drive to increase citizen participation in service provision stems not only from economic considerations, but also from a public service orientation and motivation. Local governments value citizen participation in service provision not only for enhanced efficiency and customer satisfaction, but also for democratic values such as transparency and accountability (Denhardt & Denhardt, 2000). All of these considerations have shaped the trajectory of forums for public input in service provision, culminating in the recent rise of digital platforms and mobile apps such as 311, SeeClickFix, FixMyStreet, and Street Bump.

These developments in citizen participation in service provision have been collectively described as advancements in coproduction. Coproduction, first coined by Elinor Ostrom and colleagues (Parks et al., 1981) and defined by Whitaker (1980), describes shared efforts in the joint provision of public services by government and citizens (Levine & Fisher, 1984). While most of the literature on coproduction has focused on government-to-citizen channels, recent work has also highlighted interactions amongst citizens in coproduction. Scholars have theorized about the role of groups of citizens in coproduction, differentiating between individual, group, and collective coproduction (Brudney & England, 1983; Nabatchi et al., 2017); describing community coproduction (Bovaird, 2007); and emphasizing citizen-to-citizen interactions in coproduction (Linders, 2012). Networked coproduction further recognizes that technological advances are creating digital communities through which citizens may coordinate and engage in coproduction (Meijer, 2011). This study extends this line of work.

These varieties of coproduction help us to better understand existing and new roles for citizens in service provision. Standard 311 phone call hotlines represent opportunities for individual coproduction, as an individual citizen shares information personally and unidirectionally with government. An array of digital options (e.g., the SeeClickFix app explored in this study) now also offer opportunities for interactions amongst citizens in coproduction. In particular, new service request apps allow citizens to not only learn about neighbors' requests, but also to take collaborative action on those requests by following and commenting. This interactive component differs from prior or alternative request submission processes through 311 phone call hotlines, through which neighbors' requests are not visible, and through 311 websites, through which neighbors' requests are visible but citizen-to-citizen collaboration is often not possible.

In the context of these new service request apps, I use “collaboration” to describe the relationship amongst the

citizen participants, involving commenting on and following each other's requests, and "collective citizen input" to describe the information produced by citizen collaboration using the interactive features. I suggest that the relationship amongst citizens is indeed collaborative. Citizen collaboration through service request apps occurs when citizens intentionally contribute to other citizens' requests in order to increase government attention to the issue and to thus address a shared concern. However, it is important to acknowledge that this type of collaboration frequently requires less time and effort than other forms of collective political participation, such as community organizing.

Nonetheless, a critical contribution of the collective citizen input made possible through new service request apps is that it can provide real-time information through comments and follows to validate and clarify requests. This information comes *during* the service request handling process and can thus shape how city administrators and workers address requests. Prior work on service delivery has placed theoretical and empirical emphasis on citizen coproduction and e-participation in the form of providing feedback *after* public services have been completed or measuring retrospective citizen satisfaction (Allen et al., 2020; Fitzgerald & Durant, 1980; Kelly & Swindell, 2002). In contrast, this study examines prospective, or even concurrent, "co-commissioning," as citizens submit and collaborate on service requests to direct and target where public services are then provided (Nabatchi et al., 2017). More specifically, while prior work has focused on the monitoring, tracking, and accountability features of online reporting platforms like SeeClickFix (Linders, 2012), this study highlights how these platforms enable a more active and interactive role for citizens earlier in the service delivery process. That is, citizens have opportunities to collectively shape the trajectory of issue resolution, rather than waiting until afterwards to complain if problems arise. The real-time information from multiple citizens through these interactive reporting platforms can influence city managers that are worried about errors or invalid requests, as larger "crowds" and additional contributions help to improve the quality and accuracy of information produced through crowdsourcing and coproduction (Clark, 2014; Surowiecki, 2005).

Therefore, this paper examines whether collective citizen input through coproduction impacts how city governments handle requests and how public services are delivered. As a considerable line of work has explored disparities and distributional impacts in coproduction (Clark et al., 2013; Clark et al., 2020; Jakobsen & Andersen, 2013; Xu & Tang, 2020), I'll return to an examination of the equity implications of citizen collaboration in coproduction in the Discussion. Next, I theorize that the technological advances that make citizen collaboration in coproduction possible also shape service delivery rules, building on the concepts of information accuracy and validity discussed above.

A THEORY OF COLLECTIVE CITIZEN INPUT AS A SIGNAL FOR BUREAUCRATS

Public service provision involves "service delivery rules," or standards and procedures for addressing and providing services to residents. These rules, such as a "first come, first served" rule or a procedure for addressing service requests in batches, have "distinct, definable distributional consequences which often go unrecognized" (Jones et al., 1978). I suggest that *technological advances can shape service delivery rules*. Through technological advances, such as mobile apps to submit service requests, local officials gain crowdsourced information from citizens who can affect their priorities and decisions on requests.

It is important to first consider the existing, baseline preferences and incentives of city bureaucrats and workers. Public service motivation may drive these individuals' career choices and job performance, based on rational, norm-based, or affective reasons (Perry & Wise, 1990). In addition to such pro-social, intrinsic motivations, bureaucrats often have extrinsic motivations rooted in financial incentives or the desire to receive positive feedback from others on their job performance (Ashraf et al., 2014). Relatedly, bureaucrats care about making their job performance observable and about appearing high-performing to citizens, peers, supervisors, or political overseers (Lipsky, 2010). This can lead to the prioritization of certain tasks that have the highest probability of success, clearance, or closure (Goldstein, 2022).

Technological advances might not disrupt, and may even reinforce, these incentives and existing service delivery rules. If city administrators and workers define a standard of fairness and commit to a strict "first come, first served" rule for completing service requests, then new technological platforms may further assist in defining workload queues and collective citizen input may be irrelevant. The informational gains of technological advances might also fail to affect service delivery rules if the information produced by citizens does not align with bureaucrats' incentives or address the specific informational challenges that city workers face (Buntaine et al., 2021).

In addition, although some prior work finds equitable response times across neighborhoods in cities (Clark et al., 2020), administrators may have reasons for prioritizing certain request types, political constituencies, or locations that may be immutable to technological advances and informational gains. In this case, service delivery rules would be shaped by notions of fairness and efficiency beyond those communicated through collaboration, or dictated by political principals rather than by citizen input directly (Ting, 2021). These bureaucratic incentives that define service delivery rules suggest a null hypothesis: citizen collaboration does not affect request response times.

However, there are other reasons to believe that it might. I consider a few possible explanations for how city

administrators perceive collective citizen input on requests, building on bureaucrats' baseline incentives. The perceptions of city administrators and supervisors, who are often first on the receiving end of service requests, matter for how requests are then assigned to workers and completed. When a city government receives collective citizen input on an issue, then city administrators might be incentivized to prioritize the issue for several reasons.

First, citizen collaboration can signal *validity*. In particular, administrators are more likely to believe that issues reported by multiple people represent legitimate, real issues in the community, rather than mistakes. While multiple individual requests of the same issue may also indicate validity, this can be difficult to observe across requests in the queue. Collaborative requests send an even stronger signal with followers and commenters more directly supporting and verifying an original, single request. Moreover, responding to collaborative requests helps to guard against the dominance of "frequent flyers" who might overuse or even abuse the system (White & Trump, 2018). Administrators extrinsically motivated by productivity and efficiency (i.e., not wasting time on mistakes) or intrinsically motivated by social equity would thus value signals of validity.

Second, collaboration can signal *severity* such that administrators believe that requests that garner more public attention are more serious. As prior research shows that problem severity leads to greater intergovernmental collaboration (McGuire & Silvia, 2010) and that issue salience leads to greater citizen attention and participation (Dennison, 2019; Weaver, 1991), problem severity can also lead to greater inter-citizen collaboration. That is, bigger issues are not only more likely to be reported by multiple individual citizens, but they are also more likely to be the subject of collective action. Issue severity can be conveyed through a collaborative request when the language used in comments highlights public safety risks and when the conversation between citizens echoes and reinforces shared concerns. Both the information produced through collaboration (comments) and the presence of collaboration itself (follows) can signal severity to administrators. While severity signals may indicate issues that are more difficult to resolve, they are also valuable to bureaucrats with a strong public service motivation to, for example, promote public safety.

Finally, collaborative requests can signal *scrutiny*. This channel addresses the pressure that administrators feel to resolve issues when multiple citizens are watching and waiting and builds on related research showing increased regulatory responses to public and more "visible" pollution violation reports, those with greater likes and shares, on social media (Buntaine et al., 2022). In contrast to individual request submission, collaborative requests inform citizens that their neighbors are watching and waiting for a request to be resolved as well. This can empower citizens to potentially complain together if problems arise. That is, request submission platforms with collaboration

options generate common knowledge and the potential for collective action. Importantly, city administrators know this and can avoid criticism from citizens and potential political oversight by concerned elected representatives by prioritizing collaborative requests. Thus, scrutiny signals can aid administrators concerned about their observable job performance, accountability, public upset, or potential political consequences.

In these ways, citizen collaboration provides signals that can guide where city administrators allocate time, resources, and labor in service provision. In practice, then, administrators and supervisors assigning tasks to city workers should be incentivized to implement a service delivery rule to address collaborative requests first. Therefore, I hypothesize:

Hypothesis: Government will be more responsive, measured by a higher probability of closure and shorter response times, to requests with citizen collaborators.

DATA ON SERVICE REQUEST COLLABORATION

To test whether citizen collaboration affects government priorities and speed in service provision, I use data from SeeClickFix, a service request submission app with interactive features for commenting on, following, and sharing requests. The sections below provide background information on SeeClickFix and describe the creation of the novel dataset of requests used for this study.

Background on SeeClickFix

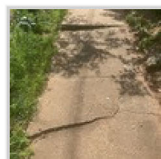
SeeClickFix is a request submission and workload management system used by more than 300 cities and towns across the US. It allows citizens to report non-emergency issues, such as potholes and graffiti, that they notice around their neighborhoods through a smartphone app or website interface. Similar to other 311 service request systems, the purpose of SeeClickFix is to "crowdsource" issue reporting. However, SeeClickFix also offers further collaboration options. In particular, SeeClickFix users not only see a map of all of the requests submitted in their neighborhood with descriptions of the issues, but they can also follow or comment on those requests.

In these ways, SeeClickFix offers opportunities for intentional citizen interaction that are not possible through individual request submission. Follows and comments on requests represent collaboration in that follows serve to validate and echo neighbors' concerns and comments often contribute additional details to support the original request. Follows and comments also serve to identify the size, and in some cases the identities, of the group of citizens involved in coproduction.

SeeClickFix



Based on issues you've acted on in the past, we felt you would be interested in this recently reported issue.



Sidewalks

715 Avery St Decatur, GA, 30030, USA

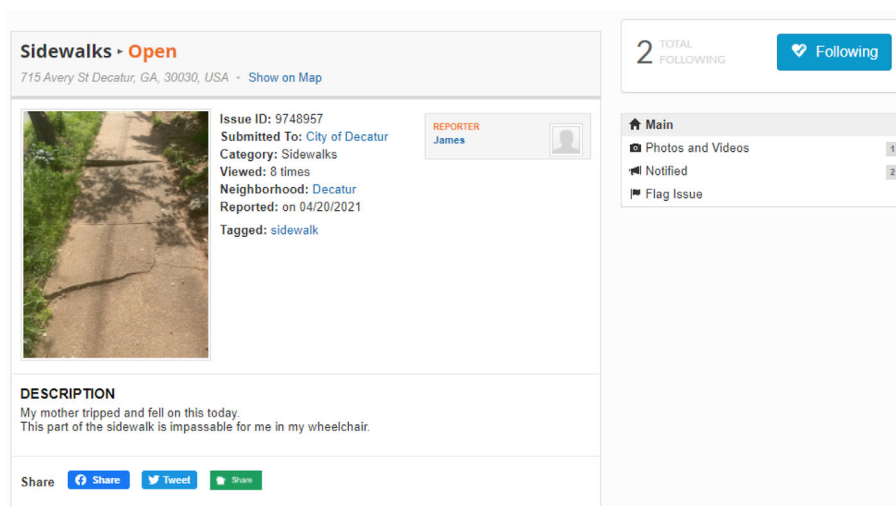
My mother tripped and fell on this today. This part of the sidewalk is impassable for me in my wheelchair. [more...](#)

Follow

Comment

Share

(a) Example Email



(b) Example Request Website

FIGURE 1 Example of collaboration on a SeeClickFix request.

Moreover, SeeClickFix sends emails to users of the app alerting them to recently reported nearby issues. The emails have direct links to follow, comment, and share the issue, encouraging citizen interaction in coproduction. The number of citizens who follow or comment on an issue is then displayed both publicly and on the back end to city administrators. Figure 1 shows an example email sent by SeeClickFix to encourage collaboration, along with the public SeeClickFix website for the request that shows the number of followers.

The data

I constructed a novel panel dataset of SeeClickFix requests for a randomly selected subset of 100 cities that

use SeeClickFix. Figure 2 shows a map of the cities in the dataset with the size of the points representing the number of requests per city. The dataset consists of over 70,000 unique requests and more than 5 million request-time period observations based on scraping request records using the SeeClickFix API for the 100 cities every 8 h between November 19, 2020 and March 18, 2021.¹ As request records scraped at a single point in time using the API only contain the current or final number of collaborators, this new panel dataset offers the opportunity to track changes in the number of collaborators over time.

To measure collaboration, the dataset includes a “rating” variable that represents the “number of unique users who have commented or followed the issue,” as expressed by SeeClickFix technical support. SeeClickFix technical support also explains that, “We define this field

Sample of Cities/Towns Using SeeClickFix

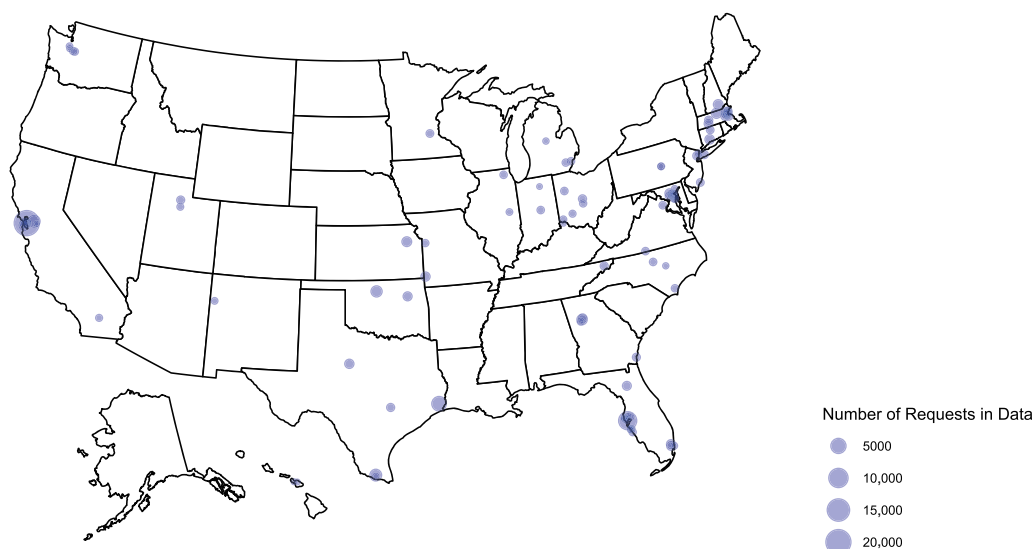


FIGURE 2 Map of cities in dataset.

in client-facing documentation as ‘The number of followers on the Request’. So, you might equate it closer to ‘issue popularity’ as opposed to ‘rate this issue’ (i.e., good/bad)” (SeeClickFix Technical Support, 2021). For reference, there is a substantial amount of collaboration on the requests in the dataset: about 40% of the requests experienced a change in rating, representing additional unique users commenting or following the issue.² I create a binary “collaboration” variable that classifies requests as collaborative if they achieve a rating score of two or higher, meaning that more than one person contributed to the request.³ Non-collaborative requests are those with a rating of only one throughout their lifespan, such that no one beyond the initial requester followed or commented on the request.

The SeeClickFix data also include information that is useful in assessing responsiveness to requests, namely the status of requests (primarily open or closed) and the times that requests are submitted and closed. I define request response time as the difference between when requests are initially submitted and when they are closed, and I measure this in hours to create a variable for the number of hours that requests remain open before closure. There are several limitations to using this response time variable. Some requests might be closed without resolution or earlier than the underlying issues are resolved if they are simply scheduled for investigation or outside city maintenance boundaries or responsibilities. Alternatively, other requests might be closed *later* than the issues are resolved if workers mark requests as completed in batches at the end of their day or shift. These issues are discussed in more detail in SI Section A.1.1.

To control for other important factors that affect the completion of service requests, I construct two additional variables using the SeeClickFix data. The first is a “workload” variable quantifying the number of open requests in a request’s city during a time period in which the request is open.⁴ Second, I create a request category variable in order to group requests that reference similar issues and that likely require similar timeframes and city resources to complete. The category variable consists of eight request categories identified through text analysis of request summaries.⁵

Finally, I additionally geolocate all SeeClickFix requests to census tracts and obtain demographic information for census tracts from the 2014–2018 American Community Survey (ACS).⁶ This demographic information by census tract—including the median age, median household income, and proportions of residents who are white, have college degrees, have internet access, have computers, commute to work, and use public transit to commute—allows me to control for factors that affect citizen collaboration on requests and that affect government responsiveness to service requests. To further consider the role of transportation and traffic conditions in the demand for and supply of services, I obtained average weekday household travel data by census tract from the Local Area Transportation Characteristics by Household (LATCH) survey by the Bureau of Transportation Statistics, and I also calculated the distance between census tract centroids and city centers as a proxy for traffic conditions and the visibility of issues in a census tract.⁷ The geolocated data include 1009 census tracts with an average of 32 requests per tract.⁸

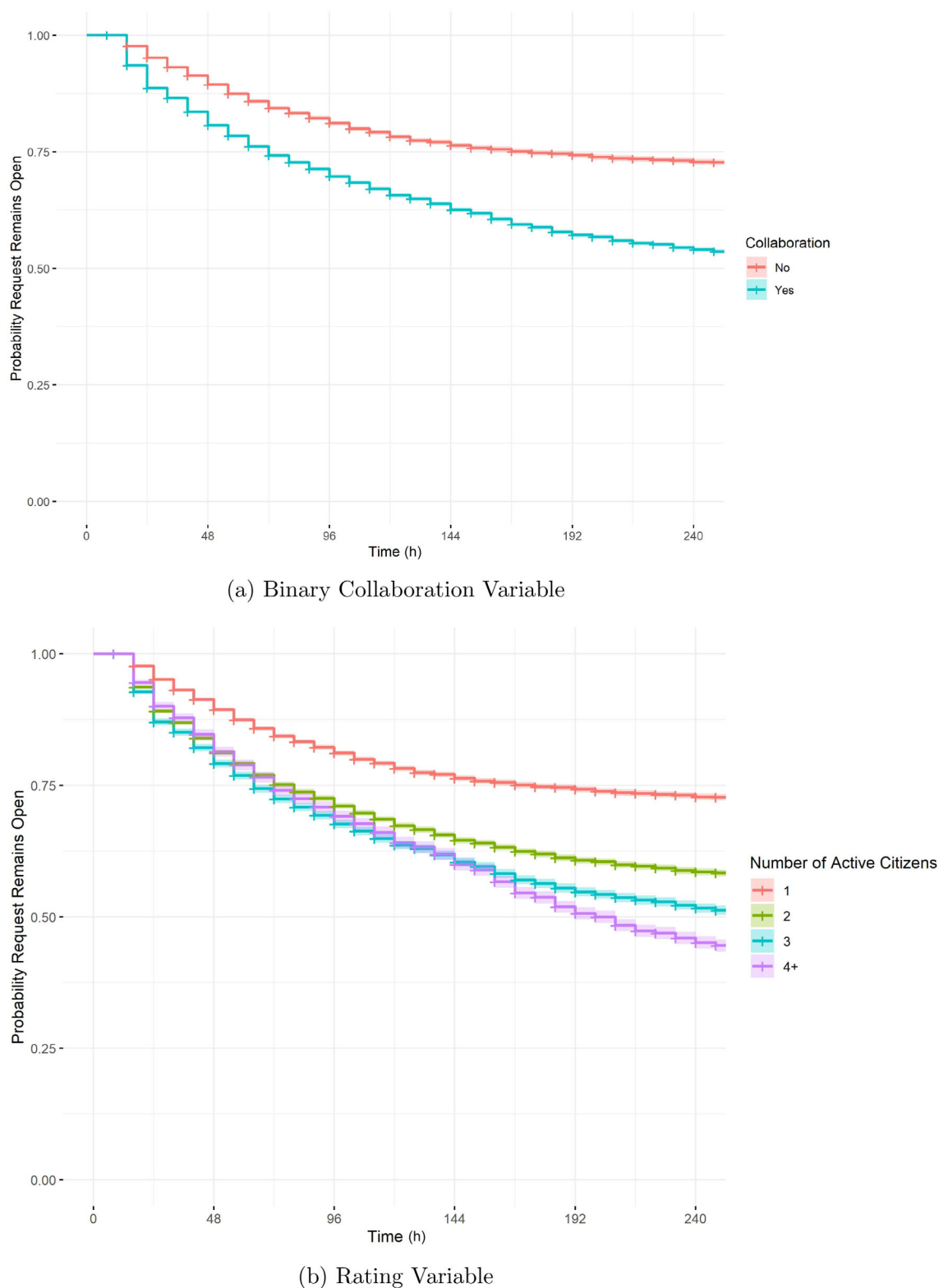


FIGURE 3 Survival probability over time for collaborative and non-collaborative requests.

ANALYSIS AND RESULTS

In order to estimate the impact of citizen collaboration on government responsiveness to service requests, I use a

survival analysis approach and a difference-in-differences approach to compare collaborative and non-collaborative requests. To examine request-level changes in collaboration and closure over time, the survival analysis approach

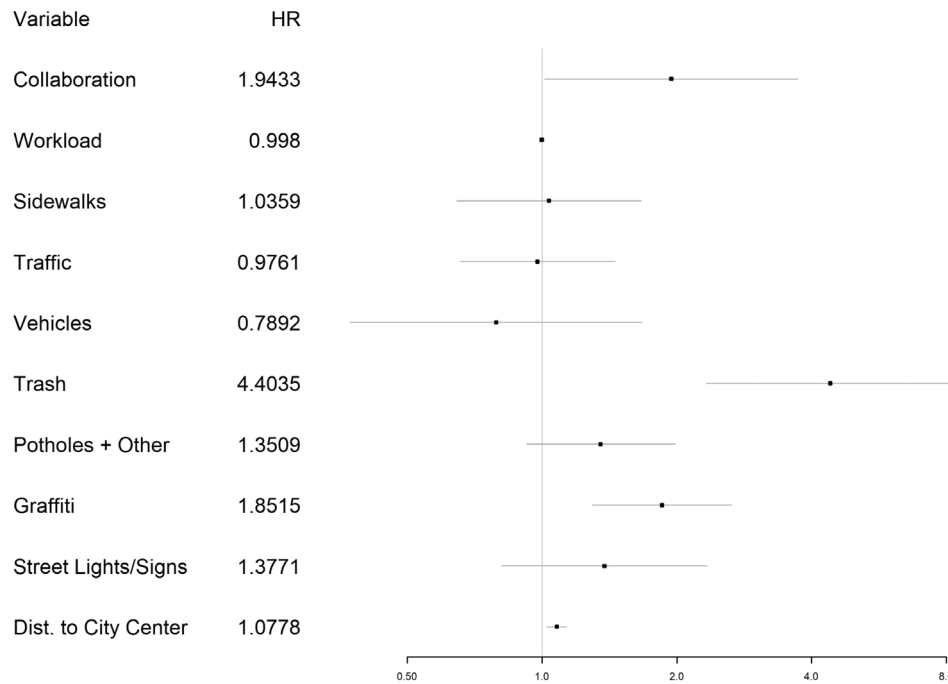


FIGURE 4 Hazard ratios (HRs) for collaboration and other variables of interest.

uses the 4 months of SeeClickFix request records scraped every 8 h for the sample of 100 cities. The difference-in-differences approach narrows in on two cities—Oakland and San Francisco—and adds in other 311 requests and SeeClickFix requests for a broader time span. For consistency, I use the binary collaboration variable in the presentation of all results, though results with the full rating variable are available in SI Section A.2. SI Section A.4 also describes methods and results for an alternative analysis approach using matching. Finally, I use interviews with city administrators to collect evidence regarding the proposed mechanisms—validity, severity, and scrutiny—driving the prioritization of collaborative requests.

Survival analysis

In order to compare the “lifespan” of service requests, Figure 3 shows Kaplan–Meier curves with 95% confidence intervals representing the probability that a request remains open over time for collaborative and non-collaborative requests. The Kaplan–Meier curve is based on non-parametric estimation of the survival function and calculates the survival probability over time for a single variable, in this case the binary collaboration variable (Efron, 1988). Figure 3a shows that the probability that a request remains open decreases more rapidly over time for collaborative compared with non-collaborative requests. For example, after 10 days (240 h), a collaborative request has about a 54% chance of remaining open, while a non-collaborative request has about a 73%

chance of remaining open. Figure 3b further shows increasing returns to collaboration: requests with more collaborators are resolved even faster. This provides some preliminary support for the hypothesis that collaborative reporting leads to improved government responsiveness in the form of faster response times on service requests. However, this method is limited to examining the bivariate relationship between collaboration and closure, and many other factors contribute to the request closure process.

Therefore, I additionally use a Cox proportional-hazards model in order to include other important covariates that affect request closure. I specifically control for request category, distance to city center, and city workload, as these variables all likely impact the time needed for cities to address requests.⁹ Figure 4 shows the results. As hypothesized, the hazard ratio (HR) for the binary collaboration variable is greater than 1 and statistically significant at the 5% level, indicating an “increased risk of death,” or in this case an increased risk of closure, for collaborative requests.¹⁰ In particular, collaboration increases the hazard, or the probability that a request is closed at a certain time given that it has not already been closed, by about 94%. For reference, the baseline hazard for a non-collaborative request after 4 days is about 20%, so a 94% increase means that a collaborative request has a hazard of about 39% after 4 days. Effectively, at each point in time after submission, collaborative requests are about twice as likely to be closed than non-collaborative requests. As for request categories, trash and graffiti requests are associated with a higher risk of closure, which comports with expectations given that they are easier to resolve than

other issues. Requests in census tracts farther away from the city center also have a slightly higher risk of closure, perhaps due to greater government responsiveness to suburban neighborhoods.

Still, a potential issue with this analysis is request heterogeneity. Collaborative requests may be different from non-collaborative requests in many ways that affect response times. In particular, collaborative requests may represent bigger problems or may be located in more visible, high-traffic areas such that these requests would be completed faster anyway. Even if we could fully account for these differences, there would still likely be other sources of unobserved heterogeneity. An option, then, is to compare request completion across cities in order to difference out consistent unobserved differences between collaborative and non-collaborative requests.

Difference-in-differences estimation

Through interviews with city administrators, I learned of a source of variation in use of SeeClickFix across two cities that offers the potential to further isolate and identify the impact of citizen collaboration on government responsiveness in service provision. In particular, I learned that while San Francisco offers SeeClickFix as a service request submission app for citizens, the city does not use the workload management interface provided by SeeClickFix to create work orders based on requests. Instead, San Francisco uses Open 311 as an alternative backend endpoint for requests.¹¹ This means that information about the comments and follows on SeeClickFix requests is not passed on to city administrators and workers in San Francisco.¹² However, the nearby city of Oakland, a prominent user of SeeClickFix, does not use Open 311 like San Francisco and instead uses the SeeClickFix workload management interface, allowing city officials to see the comments and follows on requests.¹³ Moreover, both cities have robust 311 systems that allow citizens to submit non-collaborative service requests through other means, such as phone calls, websites, and other apps.

This difference between Oakland and San Francisco creates an interesting opportunity to investigate the impact of group versus individual coproduction on government. Importantly, comparing responsiveness to collaborative and non-collaborative service requests in Oakland and San Francisco helps to alleviate some of the limitations and concerns regarding the potential for confounding and omitted variable bias in the analyses above. In particular, if collaborative requests are generally different in some important ways from non-collaborative requests, then these differences should be “differenced out” when comparing Oakland to San Francisco, leaving just differences due to city officials’ observation of collaboration. However, this approach relies on an important assumption: parallel trends. While a standard difference-in-differences approach would compare two cities over

two time periods and requires the parallel trends assumption that, in the absence of treatment, the difference in outcomes across time periods is equal in both cities, my analysis compares two cities over collaborative and non-collaborative requests and thus requires an alternative parallel trends assumption. I discuss this assumption in more detail subsequently.

In San Francisco, citizens can still follow and comment on requests, but all requests appear to government as instances of individual coproduction. In contrast, in Oakland, citizen collaboration is visible. To test whether collective citizen input on requests matters more in Oakland, where it is actually transmitted to city workers, I compile a dataset of collaborative and non-collaborative requests from 2019 to 2021 for both cities.¹⁴ The collaborative requests consist of SeeClickFix requests with a rating of two or higher. Making use of the cities’ rich open data on service requests, I incorporate non-collaborative requests from SeeClickFix (those requests without collaborators) as well as other service requests included in the cities’ 311 data.¹⁵ For 311 requests, I set the binary collaboration variable to zero, analogous to SeeClickFix requests with no collaborators. Then, I regress a response time variable for the number of hours that a request remains open on the interaction between city and collaboration, controlling for request categories and neighborhood demographics in some specifications.¹⁶ The coefficient on the interaction term represents the difference-in-differences estimate: how much collaborative requests are prioritized over non-collaborative requests in Oakland, compared with the difference in response time between collaborative and non-collaborative requests in San Francisco. The expectation is that the coefficient on the interaction term is negative, indicating that collaborative requests are closed even faster in Oakland.

Table 1 supports this expectation. Across all specifications, the coefficient on the interaction term is negative and significant, suggesting that collaborative requests are closed faster than non-collaborative requests in Oakland, compared with San Francisco. In particular, when controlling for request type and neighborhood demographics, collaboration decreases request response time in Oakland by about 156 h more, or about six and a half days more, on average than in San Francisco. In Oakland alone, request collaboration speeds up response times by about 5 days on average. This is a sizable and noticeable effect of about one work week on average, and, relative to the variability in response times for Oakland requests, represents an improvement by about 0.17 standard deviations.¹⁷

It is also evident that non-collaborative requests take longer on average to complete in Oakland than in San Francisco, giving Oakland more room to improve or speed up in resolving requests. The difference is perhaps driven by socioeconomic differences between the two cities or factors related to general government capacity and structure.¹⁸ This result is also consistent with a budget

TABLE 1 Impact of request collaboration in Oakland versus San Francisco.

	Dependent variable		
	Total hours open		
	(1)	(2)	(3)
Oakland	121.599** (1.735)	55.583** (1.805)	53.835** (1.887)
Collaboration	76.549** (3.469)	45.838** (3.499)	33.404** (3.747)
Oakland × Collaboration	−232.452** (3.898)	−169.378** (3.938)	−156.475** (4.179)
Constant	106.485** (0.285)	142.405** (0.440)	173.755** (1.457)
Observations	613,978	613,978	600,683
R ²	0.014	0.144	0.145
Request type fixed effects	No	Yes	Yes
Demographics	No	No	Yes

Note: SI Table A10 reports the full regression results with control variables. The demographic control variables, measured at the census-tract level, include the median household income, the proportion of white residents, the proportion of residents with a bachelor's degree, the number of housing units, and the proportion of households that own their residence.

* $p < .05$; ** $p < .01$.

constraint whereby faster responses to collaborative requests necessitate longer response times for non-collaborative requests. In this way, citizen-to-citizen collaboration might produce “spillover effects” across requests such that the benefits of responding to collaboration come at the cost of reduced responsiveness to non-collaborative requests or to less-collaborative neighborhoods.

Moreover, collaboration has a significant positive impact on request response times in San Francisco. This is evidence of heterogeneity between collaborative and non-collaborative requests, and, as discussed previously, could be due to issue severity. More severe issues are more likely to gain collaborators, but may also require longer amounts of time to resolve. However, this heterogeneity runs in the opposite direction than one might be concerned about for biasing results in favor of the hypothesis about the benefits of collaboration. While factors such as issue severity are not communicated through collaboration in San Francisco, Oakland city government sees collaboration on requests, can interpret collaboration as signaling severity, and can take action in response. That is, citizen collaboration more easily *identifies* the severe requests in Oakland for city workers, leading to the prioritization of these issues and resulting in faster, rather than slower, responses relative to non-collaborative requests.

Distinct request types and signals of issue severity

The parallel trends assumption in this case requires that without treatment (i.e., Oakland would not see the collective citizen input from requests, like San Francisco), the difference in response times for collaborative and non-collaborative requests would be identical between Oakland and San Francisco. One concern might be that the request

workloads, specifically the groups of collaborative and non-collaborative requests, in Oakland and San Francisco are composed of different types of requests such that differences between collaborative and non-collaborative requests between the two cities are not, in fact, parallel. While results controlling for request type are presented in Table 1, I additionally address parallel trends concerns by exploring results for more similar sets of requests.

I examine results subsetted to two distinct request categories: graffiti requests and traffic/street requests, two common high-volume request types in both cities.¹⁹ These categories additionally offer the opportunity to explore responsiveness to signals of issue severity in particular. Graffiti requests, which are not safety issues, are all arguably similarly severe, so collective citizen input may be less important for these requests. In contrast, traffic and street requests frequently represent public safety issues, so citizen collaboration can communicate the severity of issues, such as the number of people exposed to a potential safety problem.

Figure 5 reveals that the main results in Table 1 persist within the category of traffic/street requests. Oakland is faster on average in resolving collaborative traffic/street requests, but this is not the case in San Francisco. However, there is no significant difference in responsiveness to graffiti request collaboration between Oakland and San Francisco. The difference between graffiti and traffic/street requests provides some support, although not definitive, for the theorized importance of signals of issue severity.

Interviews to explore theoretical mechanisms

To further explore the proposed theoretical mechanisms—validity, severity, and scrutiny—driving the prioritization of

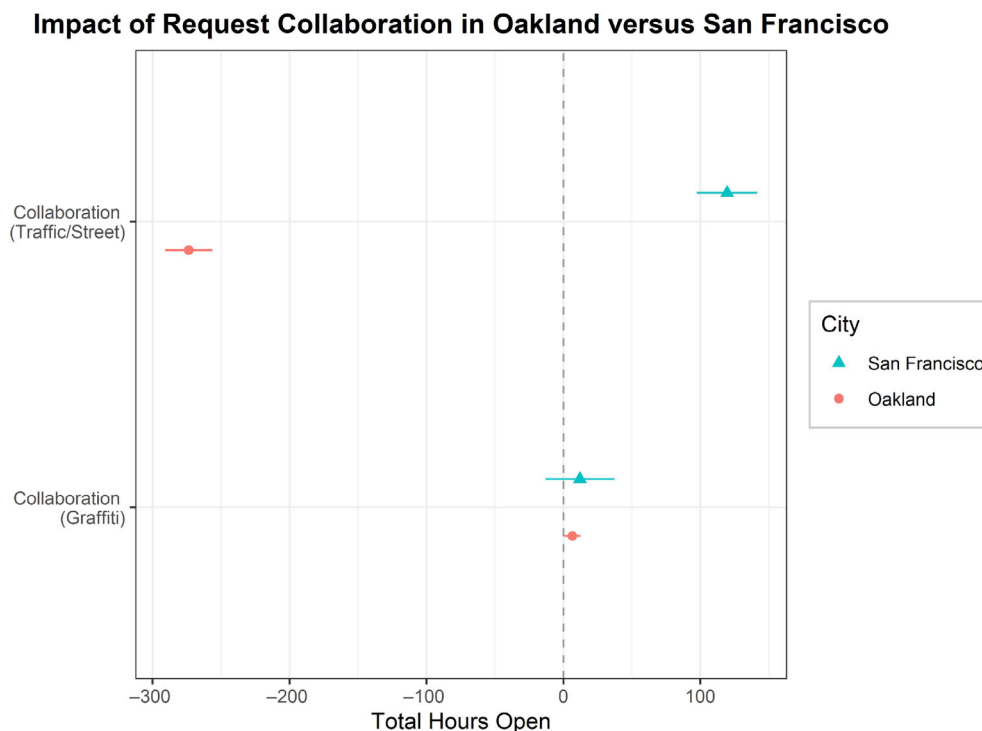


FIGURE 5 Differences in responsiveness to collaboration on graffiti versus traffic/street requests. SI Table A11 reports the corresponding regression results. SI Table A15 also presents comparisons for other request categories. The models used to produce this figure include the following demographic control variables, measured at the census-tract level: the median household income, the proportion of white residents, the proportion of residents with a bachelor's degree, the number of housing units, and the proportion of households that owns their residence.

collaborative requests, I conducted interviews in April and May 2021 with six local government officials in different cities.²⁰ In particular, I wanted to understand whether and how collective citizen input impacts how they view (and prioritize) requests. I used a semi-structured approach and asked interviewees for their thoughts on comments and follows on requests, but I did not ask questions specifically pertaining to validity, severity, or scrutiny. Instead, I examined how their responses reflected these mechanisms.

While acknowledging a default “first come, first served” rule, the interviewees also indicated responsiveness to collaboration. Importantly, almost all interviewees said that they notice comments on requests. One administrator described noticing a request for which numerous citizens added comments to complain. Another administrator explained that comments allow residents to see what others are complaining about and to see that there are other neighbors agreeing with them about issues, supporting this study's emphasis on an interactive, collective, and collaborative process. Table 2 next presents quotes and evidence from the interviews highlighting how collective citizen input signals validity, severity, and scrutiny.

Based on the interviews, there is support for all three mechanisms. Interviewees addressed validity in their explanations that collaboration weeds out unverified complaints from “habitual users” and helps to identify the “right” or “legitimate” issues to address. Referencing

TABLE 2 Evidence for mechanisms behind the prioritization of collaborative requests.

Mechanism	Evidence
Validity	- While there are some “habitual users” who might be seen more as “complainers,” when there are “others who chime in with that habitual user, it makes a difference for knowing that we have something going on.” - City workers want to address the “right” issues that are going on in neighborhoods. “Especially if it’s a legitimate issue...you know we’ll try to move a little bit quicker on that.”
Severity	Using the word “severity” directly, one administrator explained that when many people notice an issue, it helps for identifying which issues are the worst to address first.
Scrutiny	SeeClickFix allows for a “whole bunch of sets of eyes out there looking at issues.”

“severity” directly, one administrator also said that collaboration can indicate the worst issues to address first. Finally, the comment about citizen eyes watching requests supports the scrutiny mechanism. While the interviews indicate that all three mechanisms may be important, this study cannot separately identify or compare mechanisms. Exploring the distinct signals sent by citizen collaboration is a topic worthy of future study.

DISCUSSION AND CONCLUSION

The results suggest increased government responsiveness when citizens collaborate on service requests. I find that collaboration doubles the probability of request closure. Moreover, based on a difference-in-differences comparison between Oakland and San Francisco, I find that requests with comments and follows are resolved up to 5 days faster on average than non-collaborative requests. The findings suggest that citizen-to-citizen interactions in coproduction can generate information that impacts bureaucrats' job performance and that technological advances can shape service delivery rules by providing such opportunities for citizen-to-citizen interactions. Importantly, the study highlights the value of prospective and concurrent collaborative crowdsourcing for improving information accuracy and quality in service delivery (Clark, 2014). Furthermore, the findings extend scholarship examining how new technologies are changing public managers' decision-making processes and expanding collaborative governance and innovation opportunities (Kim et al., 2022).

In interpreting the difference-in-differences results, it is important to note that the fact that Oakland gathers the comments and follows on SeeClickFix requests in the first place may indicate a baseline willingness by Oakland bureaucrats to use and respond to citizen input. Therefore, the results should be interpreted as the impacts of collaboration conditional upon a government that is open to acknowledging collective citizen action. This condition may limit the generalizability of the results. However, I find largely consistent results for comparisons between San Francisco and other cities besides Oakland (see SI Section A.5.6), and recent related work arrives at similar conclusions based on an experimental study in a different country (China) and policy context (pollution violation reporting) using a different platform (Weibo) (Buntaine et al., 2022). Still, for the present study, San Francisco is the only city in the data without access to the citizen collaboration information and thus results may be driven by unique attributes of San Francisco. Future work should further explore differences across cities in adoption and implementation technology for service provision, including which platforms are selected, which functions are granted to citizen users, how collaboration is enabled (or not), and how information from citizens is used.

Another worthwhile avenue for future study concerns the functions and forms of citizen collaboration. In this study, the theorized signals sent by collective citizen input, as well as the behaviors of following and commenting, were not separately distinguishable. Yet, signals of public scrutiny and signals of issue severity may influence bureaucrats differently, especially in contexts outside of direct service provision. Moreover, following and commenting, whether online or in-person, may represent unique forms of collective citizen action. While follows, like signatures on a petition, may signal citizen solidarity, comments may serve to provide additional information and valuable on-the-ground

expertise. Future work should further explore how the inputs and outputs of citizen-to-citizen interactions in coproduction impact both citizens and governments.

How should we understand the impact of collaboration in the broader context of citizen participation in service provision and in government more broadly? Scholars of coproduction have devoted much attention to equity gaps in service provision and to the question of whether technology can elevate citizen participation to close such gaps (Clark et al., 2013; Xu & Tang, 2020). The evidence suggests that wealthier, white, and more highly-educated citizens are more likely to make requests of government and are more likely to be "frequent flyers" of service request systems, fueled by lower barriers to participation (Jones et al., 1978; White & Trump, 2018).

As a contribution to this line of research, this study highlights the potential for technology to enable more informal and less costly participation, particularly *collaborative* participation. Based on the SeeClickFix data used, there is evidence that citizens are indeed making use of new opportunities to collaborate in service provision—collaboration on the requests in my data was quite frequent, with 40% of requests gaining at least one collaborator. Furthermore, the results suggest that collaboration through echoing the concerns of neighbors and supporting existing requests of government, a potentially less information- and resource-intensive form of coproduction and political participation, can also be influential in terms of improving government responsiveness.

As technology further lowers barriers to coproduction and political participation, it is important to understand *who* participates and collaborates. On one hand, it is possible that a more diverse array of individuals may contribute due to reduced costs. However, evidence from recent work on other online forums (Einstein et al., 2022) suggests that it may be even more likely that the privileged individuals who already participate through high-cost channels will also participate through low-cost channels at a greater rate. Through collaborating, these individuals may be able to gain an even larger share of public goods and services and potentially block the distribution of goods and services to others. In addition, high levels of request submission and collaboration by city employees and by a few "frequent flyers" might further limit the democratic nature of this tool for providing input to government.²¹ Therefore, given this study's findings regarding the benefits of citizen-to-citizen collaboration in coproduction, attention should be devoted to potential equity gaps not only in individual service request submission, but also in collective service request *collaboration*, including disparities in individuals' access to support networks for coproduction.

Some cities are taking steps to address this potential issue. Through interviews with city administrators using SeeClickFix, I learned that several cities have aimed to spread awareness of SeeClickFix through efforts such as introducing the app in a "citizens' academy" and discussing the app in neighborhood improvement associations

that serve lower-income areas. While an initial investigation into the geographic distribution of collaboration on SeeClickFix requests shows no clear racial or income-based differences (see SI Section A.7), further research is needed to better account for different baseline needs and to fully understand the potential for collaboration to either close or exacerbate equity gaps in service provision. Further research should also explore the spatial dynamics, clustering, and spillover effects of request collaboration, as we know that neighbors benefit when city workers address nearby issues on their way to resolve a request (Jones et al., 1978).

If awareness of and access to interactive service request apps increase, then collaboration may create a pathway to further participation. For collaborative requests, it is possible that users who initially only comment or follow the requests of their neighbors become persuaded, conditional upon government responsiveness, to issue their own requests in the future (Sjoberg et al., 2017). Relatedly, having one's concerns echoed through a collaborative public forum may generate feelings of validation, empowerment, and self-efficacy (Meijer, 2011) such that individuals seek out additional opportunities to participate politically. However, while technology can lower the costs of participation, it may not change or expand individuals' underlying motivations or interest in politics and may not alleviate psychological costs associated with coproduction (Thomsen et al., 2020). For all of these reasons, the study raises the need to further investigate who benefits and who loses when technology is used for coproduction and citizen–government communication.

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ENDNOTES

- ¹ A potential concern is that the COVID-19 pandemic affected the types of requests submitted such that requests during this time are not representative of request submission generally. I discuss this in SI Section A.1 and present some evidence that requests submitted before and during the pandemic are similar.
- ² The SeeClickFix API does not provide the identity of collaborators, and information about the identity of initial requesters is limited.
- ³ SI Section A.2 presents results for the rating variable, which ranges from 1 to 24.
- ⁴ Department workload is less clear from this data, as the department responsible for requests is not provided by the SeeClickFix API as it is in some 311 datasets.
- ⁵ Request categories were identified through unsupervised classification of request summaries. The eight categories include issues related to sidewalks; traffic; vehicles (parked or abandoned); debris, trash, and trees; potholes; property; graffiti; and street lights and signs. See Table A1 in SI Section A.1 for more information about the categories.
- ⁶ I use the 2018 TIGER/Line shapefiles, which are available as geodatabases with the 2014–2018 ACS 5-year estimates at https://www2.census.gov/geo/tiger/TIGER_DP/2018ACS/. Clark et al. (2020) use the same data to geolocate 311 requests across 15 cities.

- ⁷ Future work could examine specific cities to get better data on road-level traffic data.
- ⁸ SI Section A.7 includes information about demographic factors associated with which census tracts submit and collaborate on requests.
- ⁹ The model specification also uses robust standard errors clustered by city, and the excluded request category is property issues.
- ¹⁰ Stratifying by, rather than controlling for, workload, request category, and distance to city center to better meet the *proportional hazards* assumption necessary for use of the Cox proportional-hazards model provides an even larger estimated hazard ratio of 3.340 for the collaboration variable, which is also statistically significant at the 1% level. SI Section A.3 presents results for additional alternative model specifications.
- ¹¹ San Francisco uses Open 311 for open data principles (transparency, open access) and also to allow citizens to submit requests through a variety of methods and platforms other than SeeClickFix.
- ¹² Excluding San Francisco from the data produces results that are consistent with, and even slightly stronger in magnitude than, the results presented for the survival analysis methods above. No other cities in the dataset use Open 311.
- ¹³ Through contacting Oakland 311 (OAK 311), I also learned that the public works department even receives direct emails when comments are added to SeeClickFix requests.
- ¹⁴ Note that this time period now includes requests submitted both before and during the COVID-19 pandemic.
- ¹⁵ I obtain 311 requests for Oakland and San Francisco through the cities' open data portals. Unfortunately, I am unable to link requests that might have been submitted through both SeeClickFix and other 311 methods.
- ¹⁶ Results are consistent when also including month dummies to account for seasonal effects. Results are also consistent, although somewhat smaller in magnitude, when including census tract by request type fixed effects for Oakland and San Francisco separately.
- ¹⁷ As shown in SI Section A.5, results are consistent across models with alternative specifications of the collaboration variable and outcome variable.
- ¹⁸ For example, San Francisco has a consolidated city-county government, potentially creating differences in workloads and service request responsibilities across the two cities.
- ¹⁹ SI Section A.1.1 discusses limitations in measuring issue resolution that might particularly affect graffiti requests.
- ²⁰ Out of 50 officials contacted, six agreed to be interviewed. For more information on these interviews, see SI Section A.1.
- ²¹ See SI Section A.6 for an investigation into whether requests are completed faster for frequent flyers.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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